AKRA ESIMISAN

Computer Science Graduate | Full-Stack & Frontend Developer | Computational Drug Discovery (Molecular Docking & Dynamics)
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PROFILE SUMMARY

Computer Science graduate from the University of Benin with hands-on experience spanning full-stack web development and computational chemistry. Builds responsive, full-featured web applications from scratch in vanilla JavaScript, HTML, and CSS, and in React, with working knowledge of Node.js, Express.js, and database-backed applications using MongoDB, PostgreSQL, and SQLite. Skilled in molecular docking and molecular dynamics simulation using AutoDock Vina, GROMACS, and CHARMM-GUI, with applied project work in computational drug discovery for Alzheimer's disease targets. Bridges both domains in MolecuLens, a deployed full-stack platform that integrates the PubChem and RCSB PDB APIs with interactive 3D molecular visualization. Combines a structured, iterative approach to problem-solving with strong self-directed learning across two technically distinct domains.

CORE COMPETENCIES

Computational Chemistry: Molecular Dynamics, Molecular Docking, Virtual Screening, GROMACS, CHARMM-GUI, AutoDock Vina, PyMOL, Mol*
Frontend: JavaScript, React, HTML, CSS, Responsive Web Design
Backend & Databases: Node.js, Express.js, REST APIs, Session-Based Authentication, MongoDB, PostgreSQL, SQLite (Turso)
Programming & Data: Python (automation, data visualization), JavaScript, Data Analysis
Tools & Platforms: Linux / WSL Ubuntu, Git, Vercel, Render, Figma
Core Skills: Technical Documentation, Problem-Solving, Self-Directed Learning

PROJECTS

MolecuLens — Full-Stack Platform for Exploring Small Molecules and Proteins
Independent Project | moleculens.vercel.app | github.com/Esimisan/MolecuLens

- Built a full-stack web app that unifies the PubChem PUG REST and RCSB Protein Data Bank (REST + GraphQL) APIs, letting users search compounds and protein structures and view identifiers (SMILES, InChI, InChIKey), computed properties, sequences, bound ligands, and SDF / PDB / mmCIF / FASTA downloads.
- Integrated the Mol* viewer for interactive 3D protein structures and ligand conformers, with a layout that adapts to its container width for phones; added side-by-side ligand-vs-ligand and protein-vs-protein comparison with differing properties highlighted and numeric deltas.
- Engineered a resilient API layer (request spacing, retry with backoff on rate-limit and server errors, in-memory caching, request cancellation) after PubChem throttling caused intermittent failures, and stored search and comparison state in the URL so every view is shareable.
- Designed an Express backend with session-cookie authentication, scrypt password hashing, and rate-limited login so users save molecules to their account; chose Turso (hosted SQLite) over a file database because Render's free tier has no persistent storage.
- Deployed the frontend on Vercel, the API on Render, and the database on Turso, with a Vercel rewrite keeping /api same-origin so cookies work without CORS.

Virtual Screening of Natural Compounds Against Acetylcholinesterase for Alzheimer's Disease
Independent Computational Research Project | github.com/Esimisan

- Screened 27 naturally occurring compounds against human Acetylcholinesterase (PDB 4EY7) using AutoDock Vina to identify candidates with potential cholinesterase inhibitory activity relevant to Alzheimer's disease.
- Automated ligand retrieval with Python and visualized the ranked binding affinities as a bar chart; identified rutin, hesperidin, naringin, sanguinarine, and evodiamine as the top-performing compounds.
- Extended the top candidate into a full molecular dynamics simulation using GROMACS and CHARMM-GUI, running system preparation and production simulation on a WSL Ubuntu Linux environment.

End-to-End Molecular Docking & Dynamics: N-Acetylglucosamine–Lysozyme Binding
Independent Computational Research Project

- Ran a complete computational pipeline from molecular docking through molecular dynamics simulation to study N-acetylglucosamine binding to lysozyme.
- Applied docking and MD workflows to characterize the protein-ligand complex, translating a static docking pose into a validated dynamic simulation.

ExpenseFlow — Full-Stack Personal Finance Tracking Application
Independent Project | expense-flow-tracker-ten.vercel.app

- Built a full-stack personal finance tracker with a vanilla HTML, CSS, and JavaScript frontend and a MongoDB database, persisting all user data on the backend for tracking income and expenses by category.
- Developed an animated doughnut chart to visualize monthly spending, giving users an at-a-glance category-level breakdown.
- Implemented budget-setting with automatic exceeded-budget warnings to help users stay within monthly targets.
- Integrated support for 80+ currencies and full dark mode across all pages, spanning the dashboard and analysis views.

Focusly — Personal Productivity Web Application
Independent Project | focusly.vercel.app

- Architected a multi-page productivity application for managing tasks, tracking goals, and running focus sessions, built entirely in vanilla HTML, CSS, and JavaScript to deepen core web fundamentals; designed a detailed Figma UI brief ahead of development.
- Engineered automatic duration-based classification logic, distinguishing short-term tasks (under one month) from long-term goals (one month or more).
- Built a persistent client-side data layer with localStorage, preserving user tasks and goals across sessions without a backend.
- Added a journey/log view and a reminder banner system with native calendar and alarm redirect links to reduce missed deadlines.

Personal Portfolio Website — React
Independent Project

- Designed and built a responsive single-page portfolio in React (Vite) with handwritten CSS, using a design-token system and relative units (rem, em, fr) so the layout adapts across screen sizes.
- Structured the site as reusable components with project data kept separate from layout, including an accessible contact form.

EDUCATION

Bachelor of Science, Computer Science
University of Benin — Benin City, Edo State, Nigeria | January 2022 – January 2025

CERTIFICATIONS & PROFESSIONAL DEVELOPMENT

AI Fluency: Framework and Foundations